Chapter 11. Bayesian Inference

Prior and posterior, Bayesian inference, Bayes and minimax estimators

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11.1 Prior and posterior distributions

In Bayesian inference the parameter is random: a **prior** density $\pi(\theta)$ expresses belief before the data, and the model $f(\mathbf{x} \mid \theta)$ is the conditional law of the data. Bayes' theorem gives the **posterior**

$$\pi(\theta \mid \mathbf{x}) = \frac{f(\mathbf{x} \mid \theta)\pi(\theta)}{\int f(\mathbf{x} \mid \theta')\pi(\theta')\,d\theta'} \;\propto\; L(\theta; \mathbf{x})\,\pi(\theta).$$

A family of priors is **conjugate** if the posterior stays in it.

Proposition 11.1 (Conjugate pairs)

(a) Bernoulli(θ) data with Beta(α_1, α_2) prior: the posterior is Beta($\alpha_1 + t, \alpha_2 + n - t$), $t = \sum x_i$.

(b) Poisson(θ) data with Gamma(α, β) prior: the posterior is Gamma($\alpha + t, \beta/(1 + n\beta)$).

(c) $N(\theta, \sigma^2)$ data (σ^2 known) with $N(\mu_0, \tau^2)$ prior: the posterior is $N(\mu_n, \tau_n^2)$ with

$$\frac{1}{\tau_n^2} = \frac{n}{\sigma^2} + \frac{1}{\tau^2}, \qquad \mu_n = \tau_n^2\Big(\frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\tau^2}\Big).$$

Proof. Multiply likelihood and prior and recognize the kernel. For (c), $-\frac{n(\bar{x}-\theta)^2}{2\sigma^2} - \frac{(\theta-\mu_0)^2}{2\tau^2}$ is quadratic in θ with leading coefficient $-\frac{1}{2\tau_n^2}$; complete the square. ■

In (c) the posterior mean is a precision-weighted average of $\bar{x}$ and μ_0, and the posterior precision is the sum of the data and prior precisions. As $n \to \infty$ the prior's influence vanishes.

Normal model with both parameters unknown. With the normal–inverse-gamma prior $\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2/\kappa_0)$ and $\nu_0\sigma_0^2/\sigma^2 \sim \chi^2(\nu_0)$, the posterior has the same form with $\kappa_n = \kappa_0 + n$, $\mu_n = (\kappa_0\mu_0 + n\bar{x})/\kappa_n$, $\nu_n = \nu_0 + n$ and $\nu_n\sigma_n^2 = \nu_0\sigma_0^2 + (n-1)s^2 + \frac{\kappa_0 n}{\kappa_n}(\bar{x} - \mu_0)^2$. Integrating out σ^2, the marginal posterior is $\frac{\mu - \mu_n}{\sigma_n/\sqrt{\kappa_n}} \sim t(\nu_n)$.

11.2 Bayesian inference

All inference uses the posterior.

- **Point estimates:** the posterior mean, median or mode.
- **Credible interval:** a set C with $P(\theta \in C \mid \mathbf{x}) = 1 - \alpha$, e.g. equal-tailed posterior quantiles. It is a probability statement about θ given the data, unlike a confidence interval.
- **Testing:** compare $P(\Omega_0 \mid \mathbf{x})$ with $P(\Omega_1 \mid \mathbf{x})$, or use the Bayes factor, the ratio of posterior odds to prior odds.

Example 11.1

With a uniform Beta(1, 1) prior and $t = 7$ successes in $n = 10$ trials, the posterior is Beta(8, 4), with mean $8/12 = 0.667$ and mode $7/10$ (the MLE). In the normal–normal case with $n = 4$, $\sigma = 2$, $\bar{x} = 3$ and prior $N(0, 1)$: the posterior precision is $4/4 + 1 = 2$, so $\tau_n^2 = \frac{1}{2}$ and $\mu_n = \frac{1}{2}(3 + 0) = 1.5$. The 95% credible interval is $1.5 \pm 1.96\sqrt{0.5} = (0.11, 2.89)$.

For large n, under regularity, the posterior is approximately $N(\hat{\theta}_{\text{MLE}}, \{nI(\hat{\theta})\}^{-1})$ (Bernstein–von Mises). So credible and confidence intervals agree asymptotically.

11.3 Bayes and minimax estimators

With a loss $L(\theta, \hat\eta)$, the **risk** is $R_{\hat\eta}(\theta) = E_\theta L(\theta, \hat\eta(\mathbf{X}))$, and the **Bayes risk** is $r(\pi, \hat\eta) = \int R_{\hat\eta}(\theta)\pi(\theta)\,d\theta$. A **Bayes estimator** minimizes the Bayes risk. A **minimax estimator** minimizes $\sup_\theta R_{\hat\eta}(\theta)$.

Theorem 11.2

The Bayes estimator minimizes the posterior expected loss $E\{L(\theta, a) \mid \mathbf{x}\}$ over a, for each $\mathbf{x}$. For squared error it is the posterior mean; for absolute error, the posterior median; for weighted squared error $w(\theta)(a-\eta)^2$, it is $E\{w\eta \mid \mathbf{x}\}/E\{w \mid \mathbf{x}\}$.

Proof. $r(\pi, \hat\eta) = \int \Big[\int L(\theta, \hat\eta(\mathbf{x}))\pi(\theta \mid \mathbf{x})\,d\theta\Big] m(\mathbf{x})\,d\mathbf{x}$ with m the marginal density of $\mathbf{X}$, so it is minimized pointwise in $\mathbf{x}$. For squared error, $E\{(a-\theta)^2 \mid \mathbf{x}\}$ is minimized at $a = E(\theta \mid \mathbf{x})$; for absolute error, at a median; the weighted case follows by differentiating in a. ■

For instance, under the loss $(\hat\sigma^2/\sigma^2 - 1)^2$ in the normal model, the Bayes estimator is $E(\sigma^{-2} \mid \mathbf{x})/E(\sigma^{-4} \mid \mathbf{x}) = \frac{\nu_n}{\nu_n+2}\sigma_n^2$, the posterior mode.

Theorem 11.3

If $\hat\eta^*$ is Bayes for some prior π and has constant risk, then $\hat\eta^*$ is minimax.

Proof. For any $\hat\eta$: $\sup_\theta R_{\hat\eta} \geq r(\pi, \hat\eta) \geq r(\pi, \hat\eta^*) = R_{\hat\eta^*}(\theta_0) = \sup_\theta R_{\hat\eta^*}$. ■

Example 11.2 (Binomial minimax)

For Beta(α_1, α_2) priors the Bayes estimator under squared error is $\frac{t+\alpha_1}{n+\alpha_1+\alpha_2}$, of the form $a\hat\theta + b$ with $\hat\theta = t/n$. Its risk is $a^2\theta(1-\theta)/n + (b + (a-1)\theta)^2$, which is constant in θ iff $\alpha_1 = \alpha_2 = \sqrt{n}/2$. Hence

$$\hat\theta^* = \frac{T + \sqrt{n}/2}{n + \sqrt{n}}, \qquad R(\theta) \equiv \frac{1}{4(1+\sqrt{n})^2},$$

is minimax. Its risk is below $\theta(1-\theta)/n$ only near $\theta = \frac{1}{2}$, so the minimax estimator is not uniformly better than $\bar X$.